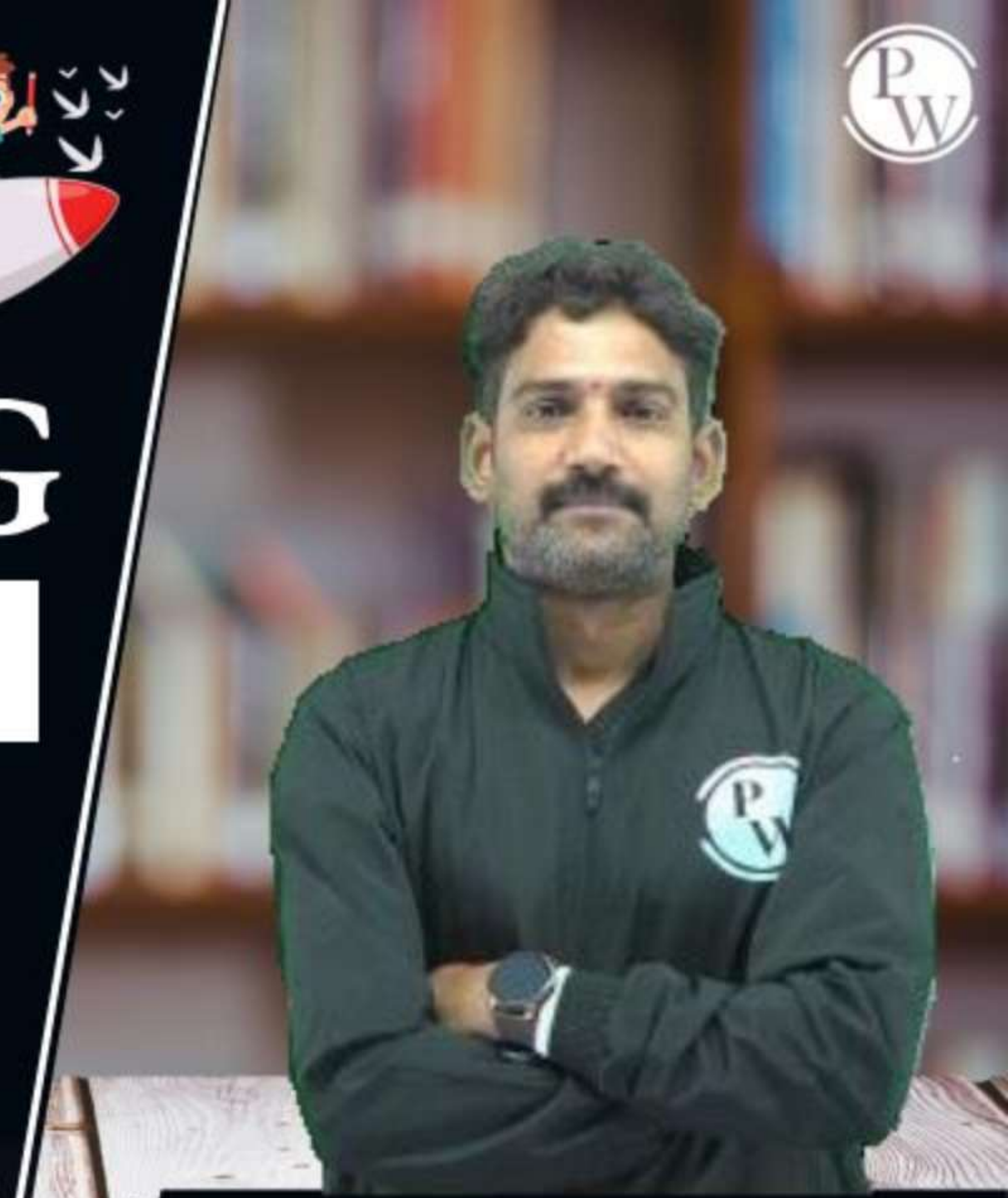
CS & IT ENGINEERING

'C' Programming

'C' TOKENS

Lecture No.- 02



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Recap of Previous Lecture

- Types of Prg languages
- Translators
- 'C' history
- 'C' features

Topics to be Covered

- Structure of a 'C' Program
- 'C' Tokens
  - Identifiers
  - Constants
  - Keywords
  - Strings

Pre-defined

'C' Library ⇒ Function Definitions, Constants.

'C' Library → Header File, Header File, ... - - - -

Ex: stdio.h

conio.h

String.h

math.h

⋮

Source code xyz.c → (compile) → Object code xyz.obj → (Execute/Run) → xyz.exe Runtime Code

College Library → CSE, DS, AI, ML

CSE → Sem-I, II, III, IV, V, VI, VII, VIII

Sem-I → Sub-1 'C', Sub-2, 3, 4, 5, 6

Topic : 'C' Tokens - 2

main( ) ⇒ Program Execution starts from main

'C' is Structured Programming Language

Title Section	// optional
Header File Section	// mandatory ✓
macro definitions	// optional
Global Declaration	// optional
main( ) function { // code }	// mandatory ✓
User-defined functions // code //Sub Program section	// optional

'C' Program

Comments

- Statements that helps for readability, understanding, Identification of code or Program.
- Comments will not be Executed but they are compiled.
- 2 Types of Comments
  1. Single line Comment // comment
  2. Multi line comment /* multi line Comment */

Inclusion of Header File

#include < Name of header file >

Example Program

Title /* Program to understand Structure and all sections of a 'C' Program */

Header File
```
#include <stdio.h>
```

Macro definition
```
#define x 10
```

Global declaration
```
int y = 5;
```

Prototype declaration
```
void fun( int);
```

main( ) function
```
int main( )
{
    int z;
    z = x + y;
    fun(z);
}
```

```
void fun( int a)
{
    Printf("%d", a*a);
}
```

User defined function

Topic : 'C' Tokens - 2

TOKENS :

- The Smallest, individual Element of a Program

- 6 Types of 'C' Tokens :

  1) Identifiers
  2) Constants
  3) Strings
  4) keywords
  5) operators
  6) Special Symbols/Seperators

Topic : 'C' Tokens - 2

{ Block
  ...... }

Example

```
int main( )
{
    int i=7, j= 8, k;
    k = i * j;
    Printf(" Result is %d", k);
    return 0;
}
```

1) int keyword
2) Space seperator
3) main Identifier
4) (operator
5)) operator
6) i Identifier
7) = operator
8) 7 Constant
9) , operator/seperator
10) j Identifier
11) 8 Constant
12) K Identifier
13) ; Seperator
14) * operator
15) Printf Identifier
16) "Result is %d" String
17) return keyword
18) 0 Constant
19) { Seperator
20) } Seperator

Topic : 'C' Tokens - 2

Identifiers

→ Name of any Programming Element →

- Variable
- function
- Constant
- Pointer
- array
- Structure
- Union
- File
- ⋮

'C' - Case-Sensitive Language

$a \neq A \qquad x \neq X$

Rules for Identifiers

1) Name should start with either an alphabet or Underscore Symbol (_)
2) No space
3) No Symbol Except underscore(_) throughout Name.
4) Name should not be a Keyword.

*5) Length of Name Should not Exceed 32 characters.

Examples

1) _x	Valid
2) x_y	Valid
3) xy_	Valid
4) @xy	Invalid
5) x@y	Invalid
6) xy+	Invalid
7) GATE	Valid
8) GATE EXAM	Invalid
9) GATE_EXAM	Valid
10) int	Invalid
11) Int	Valid
12) ___	Valid

Keywords : The words, whose meaning or Purpose is already defined.

- In 'C' Language, 32 keywords
- Also known as Reserved words.
- Each letter of keyword must be in Lower-Case.

int	signed	void	case	for	return	enum	extern
char	unsigned	if	default	break	struct	auto	sizeof
float	short	else	do	goto	union	register	const
double	long	switch	while	continue	typedef	static	volatile

CONSTANTS

- Data, whose value does not change.

CONSTANTS

- Numeric Constants [Numbers]
  - Whole Numbers
    - Without fractional Part
    - [Integer Constants]
    - Ex: 4, 19, 234, 4768 ...
  - Real numbers
    - With fractional Part (or) Precision (or) .
    - [Floating Point Constants]
    - Ex: 3.14, 0.123, 72.6794, 173.
- Literal Constants — a number (or) Symbol (or) alphabet
  - Single Literal Constants
    - Character Constants
    - Represented within ' '
  - Multi Literal Constants
    - String Constants
    - Using " "

7	⇒	Integer
7.	⇒	Float
'7'	⇒	Character
"7"	⇒	String

2 mins Summary

- Structure of 'C' Program

- Comments

- 'C' Tokens

  - Identifiers

  - Keywords

  - Constants

  - Strings

To be contd ...

THANK - YOU